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Emergency blood donation app

Innovating Emergency Blood Donation: AI, Privacy, and Efficiency in Next-Generation Apps

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Existing Apps & Systems	Real-time geolocation, AI-driven donor matching, blockchain for traceability, and privacy-preserving authentication are already in use, but nationwide integration and user engagement remain limited.	1 2 3 4 5
AI for Proximity & Prediction	AI models (gradient boosting, neural networks, federated learning) optimize donor-hospital matching and forecast demand, with privacy-preserving techniques gaining traction.	4 6
Privacy & Data Security	Blockchain, homomorphic encryption, zero-knowledge proofs, and federated learning are used to protect donor data while enabling real-time matching and analytics.	7 8
Affordability & Efficiency	IoT, cloud computing, and ride-sharing integrations improve operational efficiency, but gaps remain in accessibility, real-time updates, and donor engagement.	5 9
Research Gaps	Lack of unified, nationwide platforms combining real-time geospatial matching, advanced privacy-preserving AI, and robust donor engagement.	9 10

Direct Answer

The idea of an emergency blood request app leveraging AI for donor-hospital proximity and privacy is both timely and aligns with current technological trends. Similar solutions exist, but most lack comprehensive, nationwide integration, advanced privacy-preserving AI, and high user engagement. Building on these gaps with affordable, efficient, and privacy-focused features would represent a significant advancement in emergency blood donation systems.

Study Scope

Time Period: 2015–2024

Disciplines: Health informatics, AI/ML, blockchain, privacy engineering, emergency medicine

Methods: Systematic review, meta-analysis, comparative app analysis, synthesis of empirical and technical studies

Assumptions & Limitations

Assumes access to up-to-date, peer-reviewed literature and technical documentation.

Focuses on digital and AI-enabled solutions; does not address non-digital or purely manual systems.

Some findings are based on pilot studies or regional implementations, which may not generalize globally.

Suggested Further Research

Develop and pilot a unified, privacy-preserving, AI-driven emergency blood donation app with nationwide reach.

Investigate multi-modal data integration (IoT, blockchain, federated learning) for end-to-end automation.

Explore gamification and advanced engagement strategies to increase donor retention and trust.

1. Introduction

Efficient emergency blood donation systems are critical for saving lives, especially in trauma, surgery, and disaster scenarios. Traditional blood donation processes often suffer from delays, fragmented communication, and limited donor engagement, leading to shortages and inefficiencies. The rapid evolution of digital health technologies—particularly mobile apps, AI, and privacy-preserving frameworks—offers new opportunities to transform donor-hospital coordination, streamline emergency requests, and enhance both affordability and trust in blood supply chains.

1.1 Background and Motivation

Current challenges in emergency blood requests include:

Slow donor-recipient matching due to manual processes or fragmented digital systems.

Difficulty in recruiting and retaining new donors, especially in underserved areas.

Privacy concerns that deter potential donors from sharing sensitive health and location data.

High operational costs and inefficiencies in blood inventory management and emergency logistics.

The motivation for a dedicated app is to address these pain points by:

Leveraging AI for real-time, privacy-preserving donor-hospital proximity matching.

Automating emergency workflows and notifications.

Ensuring data security and regulatory compliance.

Making the process more affordable, accessible, and engaging for all stakeholders [1] [2] [6].

2. Landscape of Existing Blood Donation Apps and Emergency Systems

2.1 Core Features and Operational Workflows

Most current blood donation apps and emergency request platforms offer:

Real-time donor-recipient matching based on blood type and geolocation.

Push notifications to alert eligible donors within a defined radius (often 5 km) of a hospital or recipient.

User registration with health and demographic data, often verified via government or health department records.

Blockchain-based smart contracts for traceability and fraud prevention.

Donor engagement features such as digital badges, reminders, and ride-sharing integration to address transportation barriers .

Synthesis

While these features have improved response times and transparency, most platforms are regionally siloed, lack advanced privacy controls, and do not fully leverage AI for predictive analytics or personalized engagement.

2.2 Data Collection and Verification Methods

Authentication: Biometric, two-factor, and Aadhaar-based verification are used to ensure donor authenticity.

Health Data Integration: Some apps cross-validate donor eligibility with health records.

Barcode/QR Code Scanning: Used in transfusion apps to reduce errors in blood administration.

Machine Learning: Predicts repeat donation likelihood and supports targeted retention strategies [11] [12] [13].

2.3 Geolocation and Proximity Tracking

Geolocation APIs: GPS data and distance algorithms identify and alert nearby donors in real time.

Technical Challenges: Ensuring low-latency notifications (<3 seconds), accurate location updates, and dynamic donor eligibility checks.

Ride-Sharing Integration: Some platforms partner with ride-hailing services to reduce donor response times [14] [15] [16].

2.4 Blockchain and Smart Contracts in Blood Donation

Transparency: Immutable records of donations, requests, and inventory updates.

Automation: Smart contracts verify blood availability and trigger notifications.

Role-Based Access: Ensures only authorized stakeholders access sensitive data [2] [17] [18].

Synthesis

Blockchain and smart contracts are increasingly used to automate verification and inventory management, but integration with AI and real-time geospatial data remains limited in most current systems.

3. AI Integration for Donor-Hospital Proximity and Emergency Response

3.1 AI Algorithms for Real-Time Matching

GeoBlood, Rakt Setu, Blood Link: Use GPS, geospatial queries, and transformer-based recommendation systems for real-time donor-hospital matching, achieving high accuracy and rapid notifications [10] [14] [16].

Cloud-Based Systems: Enable scalable, real-time tracking and inventory management.

3.2 Forecasting Blood Demand and Donor Return

Machine Learning Models: XGBoost, LightGBM, CatBoost, LSTM, and GRU are used to forecast demand and donor return likelihood, improving supply chain resilience and reducing shortages [19] [20] [21].

Hybrid Models: Combine statistical and deep learning approaches for short- and long-term forecasting.

3.3 Privacy-Preserving AI Techniques

Federated Learning: Enables decentralized model training without sharing raw data, preserving privacy while improving model accuracy.

Differential Privacy: Adds noise to data or model updates to prevent re-identification.

Blockchain-Based AI: Ensures secure, auditable, and privacy-preserving data sharing and analytics [22] [23] [24].

3.4 Ride-Sharing and Route Planning Integrations

Dynamic Route Optimization: Reduces donor response times by integrating real-time traffic and ride-sharing data.

Spatiotemporal Flexibility: Allows delayed matching and flexible pick-up points to improve efficiency [25] [26] [27].

Synthesis

AI-driven proximity matching and demand forecasting are technically mature, but privacy-preserving deployment and seamless integration with ride-sharing and logistics remain areas for further innovation.

4. Privacy Measures in Blood Donation Apps

4.1 Anonymization and Data Masking

Techniques: (k, k^m)-anonymity, data masking, and attribute-based encryption protect donor identities.

Blockchain & Zero-Knowledge Proofs: Enable secure, pseudonymous registration and authentication [28] [29] [30].

4.2 Cryptographic Techniques: Homomorphic Encryption and Differential Privacy

Homomorphic Encryption: Allows computation on encrypted data, securing sensitive health information during analysis.

Differential Privacy: Ensures aggregate data analysis without exposing individual records [31] [32] [33].

4.3 Decentralized Privacy-Preserving Data Processing

Blockchain & Federated Learning: Support decentralized, privacy-preserving analytics and data sharing.

Permissioned Blockchains: Offer fine-grained access control and auditability [34] [35] [36].

4.4 Zero-Knowledge Proofs and Multi-Platform Authentication

ZKPs: Allow authentication and transaction validation without revealing sensitive data, supporting secure multi-platform access [37] [38] [39].

Synthesis

Advanced privacy-preserving techniques are available and increasingly adopted, but their integration into user-friendly, scalable blood donation apps is still emerging.

5. Affordability and Efficiency in Emergency Blood Requests

5.1 Blockchain for Transparency and Delay Reduction

Smart Contracts: Automate verification, inventory updates, and notifications, reducing manual delays and fraud.

Immutable Records: Enhance trust and traceability among stakeholders [40] [41] [42].

5.2 IoT Solutions for Blood Storage and Transportation

Real-Time Monitoring: IoT sensors track temperature, humidity, and location to prevent blood degradation.

RFID Integration: Enables traceability and quality control throughout the supply chain [43] [44] [45].

5.3 Cloud-Based Data Synchronization and Emergency Response

Real-Time Data Sharing: Cloud platforms synchronize donor and inventory data across hospitals, improving emergency responsiveness.

AI Integration: Enhances prediction, scheduling, and donor filtering [2] [46] [47].

5.4 Accessibility and Donor Engagement Features

Gaps: Many apps lack real-time updates, gamification, and accessibility features for older or disabled users.

Opportunities: Integrate digital badges, interactive communication, and ride-sharing to improve engagement and reduce barriers [48] [49] [50].

Synthesis

While blockchain, IoT, and cloud technologies have improved efficiency and transparency, there is a clear need for unified, accessible, and engaging platforms that address affordability and user diversity.

6. Novelty, Feasibility, and Future Directions

6.1 Comparative Analysis and Gap Identification

Feature	Existing Apps	Proposed App Potential	Gap
Real-time geolocation	Yes	Yes	Often regional, not nationwide
AI-driven matching	Yes	Yes	Limited privacy-preserving AI
Blockchain traceability	Yes	Yes	Not always integrated with AI/IoT
Privacy-preserving authentication	Partial	Yes	Advanced ZKP, federated learning underused
Donor engagement	Basic	Advanced (gamification, ride-sharing)	Lacking in most apps
Accessibility	Limited	Inclusive design	Poor support for older/disabled users

[2] [10]

6.2 Feasibility Assessment and Implementation Challenges

Technical: Integrating AI, blockchain, and IoT at scale; ensuring low-latency, high-accuracy real-time matching; maintaining interoperability.

Regulatory: Compliance with health data privacy laws (e.g., HIPAA, GDPR); managing consent and data sharing.

Operational: Stakeholder coordination, user training, and system maintenance [4] [51] [52].

6.3 Recommendations for Future App Development

Unified Platform: Develop a nationwide, interoperable app integrating AI, blockchain, IoT, and cloud technologies.

Privacy by Design: Embed federated learning, differential privacy, and ZKPs from the outset.

Gamification & Engagement: Use digital badges, interactive updates, and ride-sharing to incentivize donors.

Accessibility: Prioritize multi-language support, age-friendly interfaces, and features for users with disabilities.

Stakeholder Collaboration: Involve hospitals, blood banks, regulators, and donor organizations in co-design and governance [50] [53] [54].

Real-World Implications

A next-generation emergency blood donation app, as envisioned, could:

Reduce response times and save more lives in emergencies.

Increase donor trust and participation through robust privacy and engagement features.

Enhance transparency and accountability in blood supply chains.

Lower operational costs and improve efficiency for hospitals and blood banks.

Set new standards for privacy-preserving, AI-driven healthcare apps globally.

Future Research Directions

Pilot and evaluate a unified, privacy-preserving, AI-driven emergency blood donation app in diverse regions.

Integrate multi-modal data (IoT, blockchain, federated learning) for end-to-end automation and analytics.

Study user engagement and retention strategies, including gamification and personalized communication.

Assess regulatory and ethical frameworks for large-scale deployment of privacy-preserving health apps.

Explore interoperability with national health information exchanges and emergency response systems.

Synthesis

The proposed app idea is both viable and innovative, building on proven technologies while addressing critical gaps in integration, privacy, and user engagement. By unifying AI, blockchain, IoT, and cloud computing within a privacy-by-design framework, and by focusing on accessibility and donor motivation, such a platform could revolutionize emergency blood donation and set a new benchmark for digital health innovation [1] [4] [5] [7] [10].

References

1. How do we leverage data through replication and warehousing to meet blood collection and transfusion service needs Juskewitch, J.E., Briggs, L., Khan, J., (...), Tsang, H.C. Transfusion, 2024 https://www.scopus.com/pages/publications/85191152000?origin=scopusAI

2. Enhanced Blood Bank Transparency with Smart Contracts and Decentralized Storage on Ethereum Kaviin, P., Kavirajan, K., Subathra, G. Proceedings of the 4th International Conference on Augmented Intelligence and Sustainable Systems, ICAISS 2026, 2026 https://www.scopus.com/pages/publications/105042002850?origin=scopusAI

3. Optimizing Blood Donation Operations with a PHP-Driven Management System Tripathi, A., Kumar, A., Anand, A., (...), Paliwal, R. Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST, 2026 https://www.scopus.com/pages/publications/105014442585?origin=scopusAI

4. From prediction to practice: Barriers to implementing artificial intelligence in blood inventory management and transfusion support Kasperska-Dębowska, P., Kędziora-Kornatowska, K. Transfusion Medicine, 2026 https://www.scopus.com/pages/publications/105046442888?origin=scopusAI

5. Transforming Blood Donation Processes with Blockchain and IoT Integration: A augmented Approach to Secure and Efficient Healthcare Practices Nazir, T., Ahmed, R.H., Hussain, M., Zahid, S. 2023 International Conference on IT and Industrial Technologies, ICIT 2023, 2023 https://www.scopus.com/pages/publications/85181531640?origin=scopusAI

6. Real Time Analysis of Blood Availability in Accidental Cases using AI Singla, S., Lal, R., Negi, K.S., (...), Kumar, P. Proceedings of the 5th International Conference on Inventive Research in Computing Applications, ICIRCA 2023, 2023 https://www.scopus.com/pages/publications/85172220176?origin=scopusAI

7. Privacy-preserving techniques for big data analytics in healthcare Archana, K., Prasad, K., Ashok, M., Veerabhadram, V. Secure Big-data Analytics for Emerging Healthcare in 5G and beyond: Concepts, paradigms, and solutions, 2024 https://www.scopus.com/pages/publications/85213544161?origin=scopusAI

8. A Game Theory Approach to Preserve Privacy in Hospital Management System Arpitha, D.G. Proceedings of the International Conference on Inventive Research in Computing Applications, ICIRCA 2018, 2018 https://www.scopus.com/pages/publications/85061480838?origin=scopusAI

9. Urge to implement IOT for monitoring and preventing blood bank system crisis Pargaien, A.V., Pargaien, S., Chilwal, B., (...), Prakash, R. Journal of Medical Pharmaceutical and Allied Sciences, 2022 https://www.scopus.com/pages/publications/85133558543?origin=scopusAI

10. Blood Link: A Blood Donation Web Application with Geo Spatial Route Scheduling and Transformer Based Recommendation Systems Shanmughapriya, M., Devi, V.B., Naresh Kumar, S., (...), Praveen, S. 2025 International Conference on Computing and Communication Technologies, ICCCT 2025, 2025 https://www.scopus.com/pages/publications/105008980652?origin=scopusAI

11. D'WORLD: Blood donation app using android Meiappane, A., Logavignesh, K., Prasanna, R., Sakthivel, T. 2019 IEEE International Conference on System, Computation, Automation and Networking, ICSCAN 2019, 2019 https://www.scopus.com/pages/publications/85074751371?origin=scopusAI

12. Mobile application oriented to the attention of blood donors in the medical centers of northern Lima Sotomayor, A.J.C., Andrade-Arenas, L. Proceedings of the 2020 IEEE Engineering International Research Conference, EIRCON 2020, 2020 https://www.scopus.com/pages/publications/85097808946?origin=scopusAI

13. Data Integration Using Data Mining and SMS Reminder for Automation of Blood Donation Izadeen, G.Y., Abdulazeez, A.M., Zeebaree, D.Q., (...), Ahmed, F.Y.H. 2021 IEEE International Conference on Automatic Control and Intelligent Systems, I2CACIS 2021 - Proceedings, 2021 https://www.scopus.com/pages/publications/85112586911?origin=scopusAI

14. Real-Time Donor-Hospital Matching Using Geo-Intelligence Gugan, I., Prathishtha, S., Saranya, S., (...), Saritha, S. Proceedings ICECONF 2025 - 2025 2nd International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering, 2025 https://www.scopus.com/pages/publications/105034733387?origin=scopusAI

15. BloodConnect: Proximity Based Blood Donation Network with Real-Time Alerts and Tracking B M, N., B S, S., Samantri, A., (...), C, N. Journal of Scientific and Industrial Research, 2026 https://www.scopus.com/pages/publications/105036225877?origin=scopusAI

16. Rakt Setu - An Integrated Mobile Platform for Expediting Blood Donation and Request Management Srivastava, A., Jain, S., Sharma, P., (...), Gangadhar, T. Proceedings of 9th International Conference on

Inventive Computation Technologies, ICICT 2026, 2026
https://www.scopus.com/pages/publications/105041674807?origin=scopusAI

17. A Hybrid Blockchain-AI Architecture for Trusted Blood Donation Traceability and Availability Prediction Amdouni, I., Houaidia, C. 2026 International Wireless Communications and Mobile Computing Conference, IWCMC 2026, 2026 https://www.scopus.com/pages/publications/105044697544?origin=scopusAI

18. Block Chain Based Blood Donation Tracking System Anitha, P., Anvitha Acharya, U., Raghu, G., (...), Vaishnavi, K.S. International Conference on Emerging Technologies in Electronics and Green Energy, ICETEG 2025, 2025 https://www.scopus.com/pages/publications/105037575018?origin=scopusAI

19. The use of predictive modelling to determine the likelihood of donor return during the COVID-19 pandemic Gammon, R.R., Hindawi, S., Al-Riyami, A.Z., (...), Vermeulen, M. Transfusion Medicine, 2024
https://www.scopus.com/pages/publications/85200540308?origin=scopusAI

20. Multivariate time-series blood donation/demand forecasting for resilient supply chain management during COVID-19 pandemic Shokouhifar, M., Ranjbarimesan, M. Cleaner Logistics and Supply Chain, 2022
https://www.scopus.com/pages/publications/85135708026?origin=scopusAI

21. Machine Learning for Donor Safety and Engagement: An Analytical Assessment Raza, S., Li, N., Goel, R. Transfusion Medicine Reviews, 2026 https://www.scopus.com/pages/publications/105050429338?origin=scopusAI

22. Multi-modal federated learning with differential privacy for privacy-preserving healthcare AI Hasan, M.R., Ahmed, M.I., Saha, S., (...), Hossen, M.J. Scientific Reports, 2026
https://www.scopus.com/pages/publications/105044173794?origin=scopusAI

23. NeuroNimbus: Enabling Privacy-Preserving Real-Time Clinical Intelligence via Federated and Cloud-Orchestrated Deep Learning Govindarajan, V., Patel, P. IEEE Access, 2025
https://www.scopus.com/pages/publications/105020952457?origin=scopusAI

24. Privacy-Preserving Healthcare Data Science: A Review of Federated Learning Approaches Ammal, S.G., Sumathi, R. Proceedings of 8th International Conference on Intelligent Sustainable Systems, ICISS 2026, 2026
https://www.scopus.com/pages/publications/105036953157?origin=scopusAI

25. Tech-Enabled Blood Donation: Incorporating AI Matching, Donor Monitoring, and Ride-Sharing for Enhanced Lifesaving Efficiency Jayamari Kanna, S., Jayakumar, J., Jackulin, C., (...), Manoj, S. 2026 IEEE 18th International Conference on Computational Intelligence and Communication Networks, CICN 2026, 2026
https://www.scopus.com/pages/publications/105045431017?origin=scopusAI

26. A Distributed Model-Free Ride-Sharing Approach for Joint Matching, Pricing, and Dispatching Using Deep Reinforcement Learning Haliem, M., Mani, G., Aggarwal, V., Bhargava, B. IEEE Transactions on Intelligent Transportation Systems, 2021 https://www.scopus.com/pages/publications/85112603400?origin=scopusAI

27. Real-time optimization for relocation and dispatching of Emergency Medical Services with balanced workload and outsourced ride-hailing services Essus, Y., De La Fuente, R., Venkitasubramanian, A. Computers and Industrial Engineering, 2024 https://www.scopus.com/pages/publications/85185832254?origin=scopusAI

28. Blood and Product-Chain: Blood and its Products Supply Chain Management based on Blockchain Approach Trong, P.N., Vo, H.K., Huong, L.H., (...), Tuan, K.L. International Journal of Advanced Computer Science and Applications, 2022 https://www.scopus.com/pages/publications/85143868106?origin=scopusAI

29. Anonymizing datasets with demographics and diagnosis codes in the presence of utility constraints Poulis, G., Loukides, G., Skiadopoulos, S., Gkoulalas-Divanis, A. Journal of Biomedical Informatics, 2017
https://www.scopus.com/pages/publications/85008697494?origin=scopusAI

30. KC-Slice: A dynamic privacy-preserving data publishing technique for multisensitive attributes Onashoga, S.A., Bamiro, B.A., Akinwale, A.T., Oguntuase, J.A. Information Security Journal, 2017
https://www.scopus.com/pages/publications/85019618112?origin=scopusAI

31. Secured and cloud-based electronic health records by homomorphic encryption algorithm Annapurna, B., Geetha, G., Madhiraju, P., (...), Pandey, P. International Journal of Electrical and Computer Engineering, 2025 https://www.scopus.com/pages/publications/85209771183?origin=scopusAI

32. Artificial Intelligence (AI) Based Multi-Layered Approaches for Privacy Preservation in Federated Learning Bharath, K., Chopra, P., Kumar, M. International Journal of Modern Education and Computer Science, 2026 https://www.scopus.com/pages/publications/105036999496?origin=scopusAI

33. A Homomorphic Encryption Compiler for Blood Pressure Analysis Balaji, V., Kannan, S., Suryamritha, M., Belwal, M. 2024 3rd International Conference on Electrical, Electronics, Information and Communication Technologies, ICEEICT 2024, 2024 https://www.scopus.com/pages/publications/85208746986?origin=scopusAI

34. LifeBank: Integrating Blockchain Technology with Blood Donation Systems Yang, C., Tang, Y., Ding, Y., (...), Zhao, R. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2024 https://www.scopus.com/pages/publications/85203865937?origin=scopusAI

35. Contextual Zero-Knowledge Authentication with IPFS-Backed Hyperledger Fabric for Privacy-Preserving Blood Supply Chain Management Kamal, L., Retna Raj R, J. Applied Sciences (Switzerland), 2026 https://www.scopus.com/pages/publications/105038423087?origin=scopusAI

36. AI-DRIVEN EMERGENCY HEALTH DATA ACCESS FOR FIRST RESPONDERS USING HYBRID BLOCKCHAIN ARCHITECTURE Bheesetty, N., Alla, S., Nimmagadda, A., Park, J. Proceedings of the 2025 International Annual Conference and 46th Annual Meeting: Powering the Future of Engineering Management, ASEM 2025, 2025 https://www.scopus.com/pages/publications/105035389319?origin=scopusAI

37. Bibliometric survey on Zero-Knowledge Proof for Authentication Pathak, A., Patil, T., Pawar, S., (...), Gite, S. Library Philosophy and Practice, 2021 https://www.scopus.com/pages/publications/85103075744?origin=scopusAI

38. Research on ZKP Algorithm of Data Asset Security and Privacy Protection Based on Blockchain Technology Lan, F., Yang, J., Feng, H., (...), Chen, Y. International Journal of e-Collaboration, 2024 https://www.scopus.com/pages/publications/85200133016?origin=scopusAI

39. Blockchain Transaction Anonymity and Identity Authentication Security Based on Zero-Knowledge Proof Algorithm Xu, S. International Conference on Intelligent Communication Networks and Computational Techniques, ICICNCT 2025, 2025 https://www.scopus.com/pages/publications/105030477633?origin=scopusAI

40. Application of Blockchain Technology in Emergency Management Systems: A Bibliometric Analysis Shevchuk, R., Lishchynskyy, I., Ciura, M., (...), Kasianchuk, M. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105006748110?origin=scopusAI

41. Computer Modeling Approaches for Blockchain-Driven Supply Chain Intelligence: A Review on Enhancing Transparency, Security, and Efficiency Kumar, P.R., Mohammad, G.B., Narsimhulu, P., (...), Selvarajan, S. CMES - Computer Modeling in Engineering and Sciences, 2025 https://www.scopus.com/pages/publications/105017851514?origin=scopusAI

42. Smart Contracts, Blockchain, and Health Policies: Past, Present, and Future Kurt, K.K., Timurtaş, M., Pınar, S., (...), Türkeli, S. Information (Switzerland), 2025 https://www.scopus.com/pages/publications/105020013406?origin=scopusAI

43. The Use of Web Technology and IoT to Contribute to the Management of Blood Banks in Developing Countries Ismail, R.D., Hussein, H.A., Salih, M.M., (...), Omar, M.B. Applied System Innovation, 2022 https://www.scopus.com/pages/publications/85140410702?origin=scopusAI

44. Application of Internet of Things in clinical blood management based on RFID technology Cheng, C., Liu, S., Lv, C., (...), Sun, B. Chinese Journal of Blood Transfusion, 2021

https://www.scopus.com/pages/publications/105003473527?origin=scopusAI

45. MHUTEMP: An Open-Source IoT Solution for Reliable Real-Time Humidity and Temperature Monitoring in Healthcare Facilities Lon-Kan, C., Rodriguez, R., Lon-Kan, I. Proceedings of the 2025 IEEE 32nd International Conference on Electronics, Electrical Engineering and Computing, INTERCON 2025, 2025 https://www.scopus.com/pages/publications/105029911399?origin=scopusAI

46. Revolutionizing Blood Donor Management using Cloud Computing Partha Sarathy, E., Ramamurthy, P.M., Saxoniel, R., (...), Ramesh, A. 2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems, ADICS 2024, 2024 https://www.scopus.com/pages/publications/85195205902?origin=scopusAI

47. Real-Time Blood Donor Prediction and Campaign Optimization using DNN-Autoencoder and MERN Architecture Vikkiramapandian, D., Ajay, M., Chowmyan, R., Kumar, A.G. Proceedings - 3rd International Conference on Self Sustainable Artificial Intelligence Systems, ICSSAS 2025, 2025 https://www.scopus.com/pages/publications/105012823887?origin=scopusAI

48. Blood Resource Availability and Compatibility Locator for Real Time Management System Dalauta, C.R.S., Generalao, I.E., Simenahan, D.P., (...), Timosan, J.Q. 2025 International Conference on Innovation and Intelligence for Informatics, Computing,and Technologies, 3ICT 2025, 2025 https://www.scopus.com/pages/publications/105036343086?origin=scopusAI

49. Evaluation of the Wateen App in the Blood-Donation Process in Saudi Arabia Alessa, T. Journal of Blood Medicine, 2022 https://www.scopus.com/pages/publications/85129767141?origin=scopusAI

50. Life Line: A Location-based Mobile Blood Donation Management System Shalini, V.B., Priyan, B.P.G., Bharathi, U., (...), Amuthan, M.V. Proceedings of 4th International Conference on Self Sustainable Artificial Intelligence Systems, ICSSAS 2026, 2026 https://www.scopus.com/pages/publications/105044175073?origin=scopusAI

51. Artificial Intelligence Implementation in Transfusion Medicine: Addressing the Challenges of Clinical Adoption Maynard, S., Farrington, J., Raza, S., Stanworth, S.J. Transfusion Medicine Reviews, 2026 https://www.scopus.com/pages/publications/105030685678?origin=scopusAI

52. Harnessing the power of AI in blockchain and IoT: Challenges and solutions Garg, N., Singh, R., Kaushal, R.K., Kumar, N. AIP Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105040687051?origin=scopusAI

53. An Exploratory Study about Accessibility, Usability and User Experience with the Visually Impaired using Mobile Applications Dos S. Rosa, J.R., Valentim, N.M.C. ACM International Conference Proceeding Series, 2020 https://www.scopus.com/pages/publications/85102882807?origin=scopusAI

54. Balancing Privacy and Performance: Federated Learning with Differential Privacy for Real-Time, Resilient Healthcare AI Rahman, M.M., Yeasmin, M.S., Kabir, L., (...), Rokonuzzaman, M. Diyala Journal of Engineering Sciences, 2025 https://www.scopus.com/pages/publications/105015516125?origin=scopusAI